

# UL Unit Quiz Practice

## CS 7641: Machine Learning

### 1 Question 1 - Clustering

#### Part 1. Clustering Scenario (MCMA)

You are given a 2D dataset with the following characteristics:

- There are two elongated, curved clusters shaped like crescents (i.e., non-spherical).
- The clusters are close together at one end but clearly separated in density.

You are tasked with selecting a clustering method to analyze this data.

Which of the following statements are true in this scenario? **Select all that apply.**

- A. K-Means is likely to struggle with this dataset because it assumes clusters are spherical and uses Euclidean distance.
- B. Single linkage is likely to merge the two clusters prematurely due to chaining, especially at the close end.
- C. EM with Gaussian Mixture Models may succeed if enough components are used, as it allows overlapping densities and soft assignments.
- D. K-Means will perform well as long as the number of clusters is set to 2.
- E. Single linkage will always produce better clusters than K-Means in this scenario.
- F. Soft clustering methods like EM can assign a point near the overlapping region to both clusters with non-zero responsibility.

#### Part 2. Choosing $k$ (MCMA)

You are clustering a dataset of 500 points that visually appear to form **5 compact, roughly spherical clusters**, with some noise around the edges.

You are experimenting with different values of  $k$  in **K-Means** and **EM-based Gaussian Mixture Models (GMMs)**.

Which of the following statements are true as you vary  $k$ ? **Select all that apply.**

- A. If  $k$  is set too low (e.g.,  $k = 2$ ), both K-Means and EM are likely to merge multiple true clusters into one.

- B. If  $k$  is set too high (e.g.,  $k = 20$ ), K-Means may split a true cluster into multiple artificial parts.
- C. For very high  $k$ , the log-likelihood in EM for GMMs will generally increase, even if the clustering becomes meaningless.
- D. Choosing a high  $k$  improves interpretability, as each cluster becomes more distinct and informative.
- E. If  $k$  is too low, EM may still assign soft probabilities that capture structure better than K-Means.
- F. Both K-Means and EM are guaranteed to produce globally optimal clusterings for any fixed  $k$ .

### Part 3. Comparing EM and K-Means Clustering

You are given the following 1D dataset of points:

$$\mathcal{D} = \{1, 2, 3, 10, 11, 12\}$$

Assume the data come from two clusters. You will compare the assignments and parameter updates from **K-Means** and **Expectation-Maximization (EM)** for a Gaussian Mixture Model (GMM), using one iteration of each method.

Use the following initialization for both algorithms:

- K-Means initial centroids:  $\mu_1 = 2, \mu_2 = 11$
- EM initial parameters:
  - Means:  $\mu_1 = 2, \mu_2 = 11$
  - Shared, known variance:  $\sigma^2 = 1$
  - Mixing coefficients:  $\pi_1 = \pi_2 = 0.5$

- a) **(K-Means Assignment)** Assign each data point to the nearest centroid using Euclidean distance. Report the hard cluster assignments for all 6 data points.
- b) **(EM E-step)** For each data point  $x_i \in \mathcal{D}$ , compute the posterior probability (responsibility) that it belongs to cluster 1:

$$\gamma(z = 1 \mid x_i) = \frac{\pi_1 \cdot \phi(x_i; \mu_1, \sigma^2)}{\pi_1 \cdot \phi(x_i; \mu_1, \sigma^2) + \pi_2 \cdot \phi(x_i; \mu_2, \sigma^2)}$$

where  $\phi(x; \mu, \sigma^2)$  is the Gaussian PDF. Report all 6 values of  $\gamma(z = 1 \mid x_i)$ , rounded to 3 decimal places.

- c) (**EM M-step**) Using the responsibilities from (b), compute the updated mean  $\mu_1^{(1)}$  for cluster 1:

$$\mu_1^{(1)} = \frac{\sum_{i=1}^6 \gamma(z = 1 | x_i) \cdot x_i}{\sum_{i=1}^6 \gamma(z = 1 | x_i)}$$

Report your result rounded to 2 decimal places.

- d) (**Compare**) Briefly describe how the K-Means assignments (from a) differ from the EM responsibilities (from b). What qualitative difference do you observe in terms of hard versus soft clustering?

- e) (**New Point Classification**) Suppose a new data point  $x = 5$  is added to the dataset.

- Under the current **K-Means** centroids ( $\mu_1 = 2, \mu_2 = 11$ ), to which cluster would this point be assigned?
- Under the current **EM parameters**, compute the responsibility:

$$\gamma(z = 1 | x = 5) = \frac{\pi_1 \cdot \phi(5; \mu_1 = 2, \sigma^2 = 1)}{\pi_1 \cdot \phi(5; \mu_1 = 2, \sigma^2 = 1) + \pi_2 \cdot \phi(5; \mu_2 = 11, \sigma^2 = 1)}$$

Round your answer to 3 decimal places. Which cluster is more likely under EM?

## 2 Question 2 - Feature Selection

### Part 1. Feature Selection Methods (MCMA)

You are building a machine learning model with 200 features and want to reduce dimensionality before training. You are deciding between **filter-based** and **wrapper-based** feature selection methods.

Which of the following statements are true? **Select all that apply.**

- A. Filter methods are typically faster than wrapper methods because they evaluate each feature independently of the model.
- B. Wrapper methods tend to produce more accurate feature subsets because they directly optimize for model performance.
- C. Filter methods scale well to high-dimensional datasets because they do not involve repeated model training.
- D. Wrapper methods are guaranteed to find the optimal subset if enough cross-validation folds are used.
- E. Filter methods may discard features that are only useful in combination with others, since they evaluate features individually.
- F. Wrapper methods involve searching through subsets of features, which can become computationally expensive as the number of features grows.

### Part 2. Relevance vs. Usefulness in Feature Selection (MCMA)

You are selecting features for a classification model and evaluating both their statistical relevance to the target and their usefulness for improving model performance.

Which of the following statements are true? **Select all that apply.**

- A. A feature can be statistically relevant to the target variable but still be useless in the model if it is redundant with other features.
- B. Filtering methods rank features based on relevance, not usefulness, which may ignore interaction effects.
- C. A feature with low correlation to the target might still be useful if it improves model performance when combined with other features.
- D. Wrapping methods can capture both relevance and usefulness by directly evaluating subsets based on model accuracy.
- E. Relevance guarantees usefulness, so highly relevant features should always be included in the final model.
- F. A feature may appear irrelevant individually but become useful when combined with others in a non-linear model.

### Part 3. Feature Selection: Filtering vs Wrapping (Numerical)

You work for a moving company that wants to build a model to predict whether a customer will book a moving service after receiving an online quote (1 = books, 0 = does not book).

You have data on the following six features for each quote request:

- **Distance:** Estimated miles between origin and destination
- **ItemsCount:** Number of large household items listed
- **BookingTime:** Days between quote request and move date
- **ZipVariance:** Difference in ZIP code population density (origin vs destination)
- **HasElevator:** Whether the destination has an elevator (1 = yes, 0 = no)
- **MoveDay:** Whether the move is scheduled on a weekday (1 = yes, 0 = no)

You want to use feature selection before fitting a logistic regression model.

#### Filter-based Method (Correlation with Booking Outcome)

You calculate the Pearson correlation between each feature and whether the customer booked:

| Feature     | Correlation with Booking |
|-------------|--------------------------|
| Distance    | 0.75                     |
| ItemsCount  | 0.10                     |
| BookingTime | 0.68                     |
| ZipVariance | 0.02                     |
| HasElevator | 0.69                     |
| MoveDay     | 0.03                     |

You select the top 3 features with the highest absolute correlation.

#### Wrapper-based Method (Forward Selection with 5-Fold CV)

You evaluate different combinations of features using 5-fold cross-validation with logistic regression. The results are:

| Feature Subset                     | Accuracy |
|------------------------------------|----------|
| Distance, BookingTime, HasElevator | 0.86     |
| Distance, BookingTime, ItemsCount  | 0.83     |
| Distance, HasElevator, MoveDay     | 0.76     |
| Distance, BookingTime              | 0.79     |

### Questions

- a) Based on the filtering approach, which 3 features are selected?
- b) Suppose you mistakenly drop the third-best feature from the filter method and instead use the 4th-ranked feature (ItemsCount). According to the wrapper results, how much does the accuracy drop when using **Distance, BookingTime, ItemsCount** instead of **Distance, BookingTime, HasElevator**?

- c) Compute the marginal gain in accuracy from adding **HasElevator** to the subset **Distance, BookingTime**, based on wrapper results.
- d) Although **ItemsCount** has a low correlation with booking outcome (0.10), adding it to the subset **Distance, BookingTime** improves the model's accuracy from 0.79 to 0.83. What does this suggest about evaluating features in isolation versus in combination?

### 3 Question 3 - Feature Transformation

#### Part 1. PCA vs. ICA (MCMA)

You are working on a project to separate overlapping audio recordings from two microphones placed in different locations in a crowded room. Each microphone captures a **linear mixture** of two people speaking at the same time.

You apply both **PCA** and **ICA** to the microphone recordings in an attempt to recover the original, independent speech signals.

Which of the following statements are correct in this scenario? **Select all that apply.**

- A. PCA can reduce dimensionality and decorrelate the mixed signals, but it does not guarantee separation of independent sources.
- B. ICA is more appropriate than PCA because it assumes the original signals are statistically independent and seeks to maximize non-Gaussianity.
- C. PCA assumes the source signals are orthogonal, while ICA does not require orthogonality and can identify non-orthogonal sources.
- D. ICA can recover the original source signals up to scaling and permutation, assuming the mixing is linear and the sources are independent.
- E. PCA is suitable if the goal is to compress the data or extract directions of maximal variance, not necessarily to unmix individual sources.
- F. ICA guarantees that components are uncorrelated but not statistically independent, making it weaker than PCA for separation tasks.

#### Part 2. Overfitting in PCA and ICA (MCMA)

You are using PCA and ICA to preprocess high-dimensional data for downstream analysis. You have 100 features but only 50 samples. You plan to use the top components for visualization and clustering.

Which of the following statements correctly describe when and how PCA or ICA may overfit? **Select all that apply.**

- A. Neither PCA nor ICA can overfit, since they are unsupervised and do not use target labels during training.
- B. PCA may overfit in small-sample, high-dimensional settings by capturing noise directions with high apparent variance.
- C. ICA is more prone to overfitting than PCA when the number of components is close to the number of features or samples.
- D. Both PCA and ICA assume the input data is well-conditioned and may produce unstable components if features are highly collinear.
- E. Overfitting in PCA can be mitigated by retaining only the top  $k$  components based on variance explained.

- F. ICA may overfit by extracting spurious independent components if the true latent sources are not sufficiently non-Gaussian.

### Part 3. Comparing PCA and ICA (Numerical)

You are analyzing time-series data collected from wearable sensors placed on athletes during training. Each athlete is monitored using:

- 3-axis motion sensors (recording acceleration in the X, Y, and Z directions)
- A heart rate monitor (beats per minute)

You collect a dataset with **4 features** and **1000 samples**.

You want to perform dimensionality reduction to extract latent structure before applying a clustering algorithm to identify patterns in training intensity.

You apply both **PCA** and **ICA** to the zero-mean, standardized data matrix  $\mathbf{X} \in R^{1000 \times 4}$ , and observe the following:

#### Principal Component Analysis (PCA)

- Eigenvalues of the covariance matrix of  $\mathbf{X}$ :

$$\lambda_1 = 2.4, \quad \lambda_2 = 0.5, \quad \lambda_3 = 0.1, \quad \lambda_4 = 0.0$$

- The first principal component explains 80% of the total variance.
- You project the data onto the first two principal components before clustering.

#### Independent Component Analysis (ICA)

- ICA identifies four statistically independent components.
- The independent components are non-orthogonal but show the following excess kurtosis (i.e., kurtosis minus 3):

$$\text{kurt}(IC_1) = 1.8, \quad \text{kurt}(IC_2) = 2.1, \quad \text{kurt}(IC_3) = -0.1, \quad \text{kurt}(IC_4) = 0.0$$

- ICA projection emphasizes non-Gaussianity rather than variance.

#### Questions

- a) **PCA Variance Retention.** Compute the percentage of total variance retained when projecting onto the first two principal components. Show your computation.
- b) **PCA Objective.** Compute the total variance in the original dataset and explain why PCA ranks the components as it does.

- c) **ICA Non-Gaussianity.** Compute the total excess kurtosis (i.e., total non-Gaussianity) across ICA components, and explain how ICA selects components differently than PCA.
- d) **Interpretation.** Suppose clustering with PCA leads to overlapping groupings, while ICA reveals distinct clusters aligned with different heart rate zones. What does this suggest about the latent structure of the data?
- e) **Method Selection.** If your goal is to recover independent physiological and movement signals from the mixed sensor data, which method would be more appropriate (PCA or ICA) and why?

## 4 Question 4 - Manifold Learning

You have a dataset of professional luchadores with mixed numerical and categorical features: career win rate, average bout tempo, signature-move diversity, rivalry graph metrics (degree, betweenness), weight class, mask style, and promotion/arena history. Your goal is to produce a 2D embedding for exploratory scouting and match-ups.

### Part 1. Compare t-SNE and UMAP (MCMA)

**Which of the following statements about t-SNE and UMAP are correct in this setting? Select all that apply.**

- A. In t-SNE, the *perplexity* parameter acts like an effective neighborhood size, balancing local versus semi-global structure.
- B. In UMAP, increasing *n\_neighbors* tends to emphasize broader, more global structure at the cost of very local detail.
- C. t-SNE optimizes a cross-entropy on fuzzy simplicial sets built from a *k*-NN graph.
- D. UMAP builds a (approximate) *k*-NN graph and optimizes a cross-entropy between high- and low-dimensional fuzzy simplicial sets.
- E. By design, t-SNE preserves global pairwise distances better than UMAP on large, heterogeneous datasets.
- F. Decreasing UMAP's *min\_dist* (e.g., toward 0) typically yields tighter, more compact clusters with larger gaps between them.

### Part 2. Workflow Choices (MCMA)

You plan to: (i) embed 50,000 current luchadores; (ii) later map in 5,000 new entrants from an upcoming tournament; (iii) compare styles while respecting non-Euclidean similarities (e.g., graph distances on the rivalry network); and (iv) avoid misleading density artifacts when interpreting cluster areas.

**Which statements are correct for these goals? Select all that apply.**

- A. UMAP supports adding new points to an existing embedding (**transform/update**) without refitting the entire model; vanilla t-SNE does not, unless you use a parametric or specialized variant.
- B. Out of the box, both methods equally support a wide range of distance metrics (including many non-Euclidean ones) to the same extent.
- C. t-SNE embeddings tend to preserve cluster *densities/areas* proportionally to original sample densities, so blob size is a reliable proxy for class prevalence.
- D. UMAP has a supervised mode that can use label information (e.g., weight class or promotion) to shape neighborhoods while still producing a 2D embedding.

- E. For very large datasets (hundreds of thousands of rows), UMAP typically scales better in runtime and memory than standard t-SNE implementations.
- F. UMAP uses an *early exaggeration* phase to inflate attractive forces at the start of optimization.